

- Q.27 Explain Einthoven's triangle with diagram.
 Q.28 Write a short note on apnea monitor
 Q.29 Explain Lithotripter with block diagram.
 Q.30 Write a note on noise in medical instrumentation.
 Q.31 Explain different body potential & explain generation of bio-potential in body.
 Q.32 Write a note on Biomedical simulator.
 Q.33 Explain muscular system and its functions.
 Q.34 Write a short note on blood sugar measurement.
 Q.35 Explain bed side monitor with diagram.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain ECG machine with block diagram.
 Q.37 Explain Heart lung machine with diagram.
 Q.38 Write note on following :
 a) Audiometer
 b) ophthalmoscope

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3rd Sem / Med Eltx

**Subject:- Biomedical Instrumentation- 1 / Biomed. Instr./
 Basic of Instr.**

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What is necessary for providing a common reference measurement
 a) Active Electrode b) Ground Electrode
 c) Tape Recorder d) Oscilloscope
- Q.2 The filter used for prevention of Noise interference
 a) High pass b) Low pass
 c) Notch d) None of the above
- Q.3 Which of the following is a stress testing device
 a) Patient Monitor b) X-ray machine
 c) Tread Mill Test d) All of the above
- Q.4 Tidal volume concern with
 a) X-Ray Machine b) ECG
 c) Ventilator d) Pulse oximeter
- Q.5 How many electrodes placed in 10/20 electrode placement system

- a) 19 b) 20
- c) 21 d) 10

Q.6 Equipment History card are meant for

- a) The way equipment behave
- b) Total down time
- c) Wear off time of component
- d) All

Q.7 Residual volume concern for

- a) Heart b) Lungs
- c) Liver d) None of the above

Q.8 CT stands for

- a) Controlled tomography
- b) Comparison tomography
- c) Converted tomography
- d) Computerized tomography

Q.9 LED stands for

- a) Light Emitting display
- b) Low Emitting display
- c) Light Emitting detector
- d) Light Emitting diode

Q.10 One cardiac cycle takes

- a) 1.5 sec b) 0.5 sec
- c) 0.8 sec d) None of the above

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Hypertension

Q.12 Pulse oximetry

Q.13 Diastole

Q.14 Stroke volume

Q.15 Montage

Q.16 Expand ICU

Q.17 What is the Normal Blood pressure of a healthy person?

Q.18 AV Node.

Q.19 Endoscopy

Q.20 What is Residual Volume?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Discuss any two types of hearing aid.

Q.22 Write a short note on ambulatory monitoring system.

Q.23 Explain central monitoring system with diagram.

Q.24 Explain mechanics of breathing with diagram.

Q.25 Explain EEG with block diagram.

Q.26 Write a note on Oxygen concentrator

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